PAPER_554: Buoyancy-Stratified Factorial Geometry — Riemann Curvature of the Aether Metric

Unified Quantum Field Framework (UQFF)

Daniel T. Murphy • UQFF Research Consortium • 2026

Source: PAPER_554_BSFG_Riemann_Curvature_Aether_Metric.md

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Author: Daniel T. Murphy — Star Magic / UQFF Framework **Session:** 148 | **Source:** Composed from CP4 #43, #66, #67 (Sessions 107–110) **CP4 Class:** `BSFGRiemannCurvatureAetherMetricCalculator` (#149) **Date:** 2026-03-27

> ***Context note:*** *The Aether metric $A_{\mu\nu}$ was introduced in PAPER_392 (CP4 #43) with coupling $\eta = 10^{-22}$ and scalar amplitude T_{s00}. PAPER_416 (CP4 #66) provided the five-component spatial decomposition of $T_{s00}(r)$. This paper combines those results to derive for the first time the full Riemann curvature tensor of the BSFG geometry — the first paper in the BSFG Complete Geometric System series (PAPER_554–558).*

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§1 Abstract

The Buoyancy-Stratified Factorial Geometry (BSFG) is defined on a 26-dimensional pseudo-Riemannian manifold $\mathcal{M}^{26}$ equipped with the Aether-perturbed metric $A_{\mu\nu}(r) = g_{\mu\nu} + \varepsilon(r)\,\delta_{\mu\nu}$, where $\varepsilon(r) = \eta\,T_{s00}(r)\,\cos(\pi t_n)$ and $T_{s00}(r)$ is the five-component stellar stress-energy density from PAPER_416. This paper derives the Christoffel symbols, Riemann curvature tensor, Ricci tensor, and Ricci scalar for the 4D slice of BSFG at a field point r. The key result is:

$$R^r{}_{0r0} \approx \frac{\varepsilon''}{2} = \frac{6\eta\cos(\pi t_n)\,C_{\rm num}}{r^5}$$

with $C_{\rm num} = (M_s c^2 + L_s/c^2)/(4\pi/3)$. At the solar surface $r = R_\odot$: $R^r{}_{0r0} \approx 1.56 \times 10^{-19}\ {\rm m}^{-2}$, which is $\approx 4 \times 10^{-26}$ times the Schwarzschild curvature — confirming the Aether contribution is a genuine but ultra-weak geometric perturbation.

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§2 The Aether Metric — Previous Results

From PAPER_392 (CP4 #43), the BSFG metric is:

$$A_{\mu\nu} = g_{\mu\nu} + \eta\,T_{s00}\,\cos(\pi t_n)\,\delta_{\mu\nu}$$

In components (4D Minkowski background with signature $+{-}{-}{-}$):

$$A_{\mu\nu} = \mathrm{diag}(1+\varepsilon,\,-1+\varepsilon,\,-1+\varepsilon,\,-1+\varepsilon)$$

where $\varepsilon = \eta\,T_{s00}(r)\,\cos(\pi t_n)$. From PAPER_416 (CP4 #66), the five-component Ts00 with dominant radial term:

$$T_{s00}(r) = \underbrace{\frac{M_s c^2}{\frac{4}{3}\pi r^3}}_{T_1 \propto r^{-3}} + \underbrace{\frac{L_s}{c^2\cdot\frac{4}{3}\pi r^3}}_{T_2 \propto r^{-3}} + \underbrace{\frac{\rho_{SCm}v_{SCm}^2}{c^2} + \frac{\rho_A v_{UA}^2}{c^2} + \rho_{sw}v_{sw}^2}_{\rm constant\ terms}$$

The radially-dependent numerator is $C_{\rm num} = (M_s c^2 + L_s/c^2)/(4\pi/3) \approx 4.27 \times 10^{46}\ {\rm J\cdot m}$.

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§3 Christoffel Symbols

Step 1. Compute $\varepsilon(r)$ and its radial derivatives:

$$\varepsilon'(r) = \frac{d\varepsilon}{dr} = -\frac{3\eta\cos(\pi t_n)\,C_{\rm num}}{r^4}, \qquad \varepsilon''(r) = +\frac{12\eta\cos(\pi t_n)\,C_{\rm num}}{r^5}$$

Step 2. For a diagonal metric $A_{\mu\mu}(r)$ depending only on $r = x^1$, the non-zero Christoffel symbols of the Levi-Civita connection are:

$$\Gamma^r_{\mu\mu} = -\frac{\partial_r A_{\mu\mu}}{2\,A_{rr}} = -\frac{\varepsilon'}{2(-1+\varepsilon)} \qquad \text{(no sum on }\mu\text{)}$$

$$\Gamma^\alpha_{\alpha r} = \Gamma^\alpha_{r\alpha} = \frac{\partial_r A_{\alpha\alpha}}{2\,A_{\alpha\alpha}} \qquad \text{(no sum on }\alpha\text{)}$$

Explicitly at leading order in ε:

Symbol	Exact	Leading order
Γ^r_{00}	$-\varepsilon'/(2(-1+\varepsilon))$	$+\varepsilon'/2$
Γ^r_{rr}	$+\varepsilon'/(2(-1+\varepsilon))$	$-\varepsilon'/2$
Γ^0_{0r}	$+\varepsilon'/(2(1+\varepsilon))$	$+\varepsilon'/2$
Γ^i_{ir}	$+\varepsilon'/(2(-1+\varepsilon))$	$-\varepsilon'/2$
Γ^r_{ii}	$-\varepsilon'/(2(-1+\varepsilon))$	$+\varepsilon'/2$

§4 Riemann Curvature Tensor

Step 3. Apply the Riemann tensor formula:

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu\Gamma^\rho_{\nu\sigma} - \partial_\nu\Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}$$

The dominant component (tidal force in the radial-temporal plane):

$$R^r{}_{0r0} = \partial_r\Gamma^r_{00} + \Gamma^r_{rr}\Gamma^r_{00} - \Gamma^r_{00}\Gamma^0_{0r}$$

$$= \frac{\varepsilon''}{2} - \frac{(\varepsilon')^2}{2} + O(\varepsilon^3) \approx \frac{\varepsilon''}{2}$$

Step 4. Substituting $\varepsilon'' = 12\eta\cos(\pi t_n)C_{\rm num}/r^5$:

$$\boxed{R^r{}_{0r0} = \frac{6\eta\cos(\pi t_n)\,C_{\rm num}}{r^5}}$$

The higher-order $(\varepsilon')^2$ correction is of order $\eta^2\,T_{s00}^2 \sim 10^{-30}$, negligible.

§5 Ricci Tensor and Ricci Scalar

Ricci tensor — applying $R_{\mu\nu} = R^\rho{}_{\mu\rho\nu}$ with $SO(3)$ spherical symmetry ($x^2 = y$, $x^3 = z$ contribute equally to $x^1 = r$):

$$R_{00} = R^r{}_{0r0} + R^\theta{}_{0\theta 0} + R^\phi{}_{0\phi 0} = 3\,R^r{}_{0r0}$$

$$R_{rr} = -R^r{}_{0r0} + 2\!\left(\frac{\varepsilon''}{2} - \frac{(\varepsilon')^2}{4}\right)$$

Ricci scalar:

$$R = A^{\mu\nu}R_{\mu\nu} = \frac{R_{00}}{A_{00}} + \frac{R_{rr}}{A_{rr}} + \frac{2R_{\theta\theta}}{A_{\theta\theta}}$$

Kretschner scalar (leading order):

$$K = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} \approx 12\,(R^r{}_{0r0})^2$$

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§6 Numerical Values at the Solar Surface

At $r = R_\odot = 6.96 \times 10^8\ {\rm m}$, $t_n = 0$ (maximum coupling), $\eta = 10^{-22}$:

Quantity	Value	Notes
$\varepsilon(R_\odot)$	1.27×10^{-2}	Not small — linearization is qualitative at surface
$\varepsilon'(R_\odot)$	$-5.47 \times 10^{-11}\ {\rm m}^{-1}$	Aether gradient
$\varepsilon''(R_\odot)$	$+3.13 \times 10^{-19}\ {\rm m}^{-2}$	Curvature driver
$R^r{}_{0r0}$	$+1.56 \times 10^{-19}\ {\rm m}^{-2}$	BSFG tidal curvature
$R_{\rm scalar}$	$\approx +3.0 \times 10^{-19}\ {\rm m}^{-2}$	de Sitter (gravity-dominant)
K	$\approx 2.9 \times 10^{-37}\ {\rm m}^{-4}$	Kretschner scalar

For comparison, the Schwarzschild curvature at $R_\odot$: $R^r{}_{0r0}|_{\rm GR} \approx GM_\odot/R_\odot^3 \approx 3.95 \times 10^{-7}\ {\rm m}^{-2}$, giving:

$$\frac{R^r{}_{0r0}|_{\rm BSFG}}{R^r{}_{0r0}|_{\rm GR}} \approx \frac{1.56 \times 10^{-19}}{3.95 \times 10^{-7}} \approx 3.9 \times 10^{-13}$$

The BSFG Aether curvature is $\sim 10^{12}$ times weaker than GR at the solar surface, consistent with the perturbative assumption $\eta \sim 10^{-22}$ being a tiny coupling.

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§7 Connection to UQFF Framework

The Riemann tensor $R^r{}_{0r0}$ of the BSFG geometry provides the **geometric encoding of tidal UQFF forces**. The buoyancy force F_U^{bi} represents the differential curvature between interior ($r < R_*$) and exterior ($r > R_*$) regions — a curvature discontinuity at the stellar boundary. The Aether field $\varepsilon(r) \propto r^{-3}$ creates a genuine non-flat geometry whose curvature $\propto r^{-5}$ decays rapidly away from the source, consistent with the UQFF fifth-force measurements reported in PAPER_413–418.

Hub: PAPER_554 (#149) is the first paper in the BSFG series. See PAPER_558 (#153) for the complete geometric system definition and PAPER_555 (#150) for metric compatibility and geodesic equation.